Physics – Chapter 11

"Work – Energy – Power"

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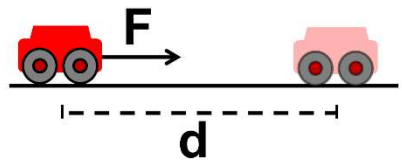
WORK

- If a constant force F is exerted on an object which moves through a displacement $\vec{d}$ parallel and of same direction to $\vec{F}$, then:

Work done = force applied x distance moved or w = F x d



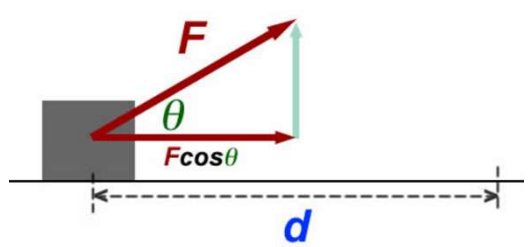
The SI unit of force is (N) and for distance (m), as a result the work's unit is (N.m) which is abbreviated (J);

1 newton-meter = 1 joule.

App1. A 10 N force F pulls on a cart through a displacement d of 5 m. How much work was done by this force?

App2. A 20 kg suitcase is raised 3 m above a platform by a conveyor belt. How much work is done on the suitcase? Use: g = 10 m/s^2.

- However, if the angle (θ) between F and d isn't zero, then: **W = F x d x cos (θ)**

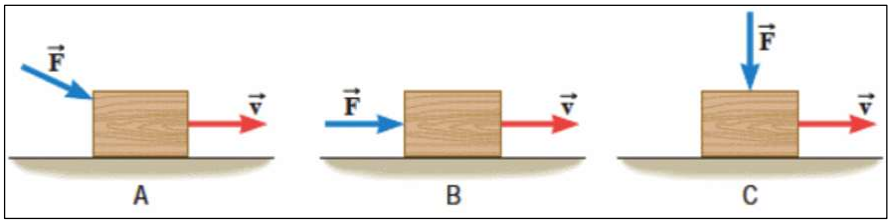


App3. How much work is done on a vacuum cleaner pulled 3 m by a force of 50 N at an angle of 60° above the horizontal?

- Depending on the angle (θ), an applied force can either cause an object to slow down (negative work if θobtuse), or speed up (positive work if θ acute). When work is positive it said to be motive. However, a negative work is called resistive.

- Note: $w = F.d.\cos(\theta) \Rightarrow w = \vec{F} \cdot \vec{d}$

App4. The same force ***F*** pushes in three different ways on a box as the drawings below show. Rank the work done by the force ***F*** in ascending order (smallest first):



A. (A, B, C) **B.** (A, C, B) **C.** (B, A, C) **D.** (C, B, A) **E.** (C, A, B)

Resultant Work (Total Work)

It is the sum of the work of all forces acting on an object or a system.

App5. A worker pushes a 150 kg crate with a horizontal force of 345 N a distance of 24 m. Assume that the frictional force between the crate and the floor is 330 N.

a) Calculate the work done by the worker on the crate.

b) For the following forces acting on the crate, frictional force, ground normal reaction, and earth's gravitational pull, how much work is done by each of them?

c) Finally, deduce the total work done on the crate? Interpret the result found.

ENERGY

- To do work on an object energy is used (Don't forget, energy is ability to work). Also, work transfers energy to the object receiving this work.
- Energy exists in many forms; regardless the form of energy, its SI unit is joules (J).

Kinetic Energy

The energy of an object that is due to its motion is called kinetic energy:

Kinetic energy = $\frac{1}{2}$ x mass x speed squared or KE = $\frac{1}{2}$. m . v^2

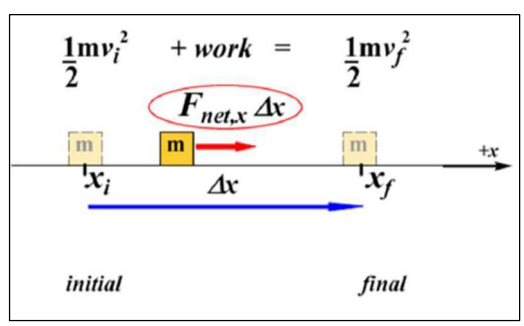
App6. A 6 kg cat runs after a mouse at 10 m/s. Calculate the cat's kinetic energy?

App7. What is the speed of a 0.145 kg baseball if its kinetic energy is 109 J?

Work–kinetic energy theorem

- The total work done by all the external forces acting on an object is equal to the change in the object's kinetic energy:

Total work = change in kinetic energy or $\sum W_{\vec{F}_{ext}}$ = ΔKE = KE_f - KE_i

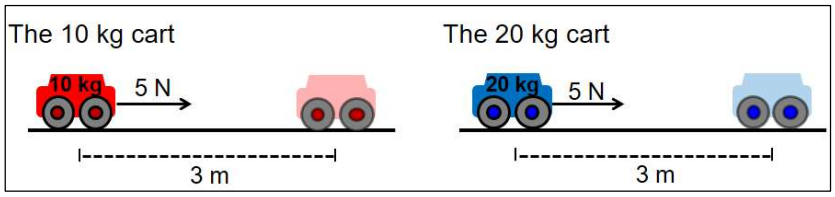


3 cases are present:

- If resultant work is (+) or motive ⇒ $\Delta KE > 0 \Rightarrow KE_f > KE_i \Rightarrow v \nearrow$
- If resultant work is (-) or resistive ⇒ $\Delta KE > 0 \Rightarrow KE_f > KE_i \Rightarrow v \searrow$
- If resultant work equals zero ⇒ $\Delta KE = 0 \Rightarrow KE_f = KE_i \Rightarrow v = constant$

- *In one word: work varies an object's kinetic energy.*

App8. A 10 kg cart and a 20 kg cart start from rest on a low-friction surface. Each cart is pulled to the right with an external net force of 5 N over 3 m as shown below.



a) On which cart is the most work done?

b) Which cart gains the most kinetic energy?

c) Which cart has the greatest final speed after moving the 3 m?

App9. A student wearing frictionless skates on a horizontal smooth surface is pushed by a friend with a constant 45 N-force. How far must the student be pushed, starting from rest, so that his final kinetic energy is 352 J?

App10. Starting from rest, a 75 kg bobsled is pushed along a horizontal surface by some athletes. Find:

a) The magnitude of the net force on the sled if it reaches a final speed of 6 m/s when pushed a distance of 4.5 m.

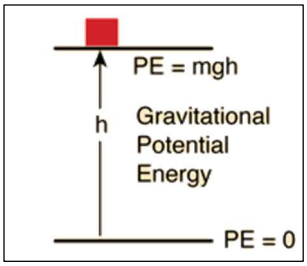
b) The distance the sled should be pushed through to reach a final speed of 9 m/s using the same force above.

Potential Energy (Stored Energy)

It is the energy stored in a system due to object's position or object's shape.

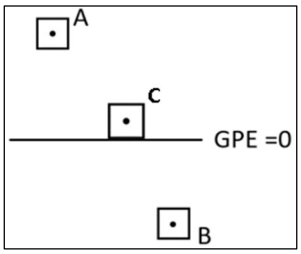
Gravitational potential energy

- As an object is moved against gravity the work done on the object is stored as gravitational potential energy in the system (object ; Earth). As result:



Gravitational potential energy = mass x free-fall acceleration x height or
GPE = m.g.h

- The height h is measured relative to a zero level which is often taken the floor or sometimes sea level.



$GPE_A = +mgh_A$ (since A is above the reference)

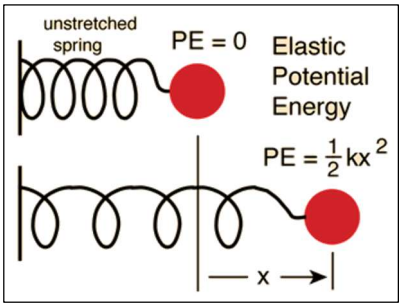
$GPE_B = -mgh_B$ (since B is below the reference)

$GPE_C = 0$ (since C belongs to the reference of GPE)

App11. Calculate the gravitational potential energy of a 1.5-kg book placed on a 2 m high shelf.

- Note:
  $w_{m.\vec{g}} = m.g.(h_i - h_f) = -\Delta GPE$

Elastic potential energy of a spring



- Elastic potential energy is stored in any compressed or stretched object, such as a spring, stretched rubber bands, bent surface of a trampoline ….

  Elastic potential energy = $\frac{1}{2}$ x spring constant x (distance compressed or stretched)^2 or EPE = $\frac{1}{2}$. k . x^2

- The spring constant k which is measured in N/m, tells how hard or soft the spring it is.

App12. If the spring constant is 100 N/m, then what is the potential energy of the spring if it is stretched 0.2 meters from its original length?

App13. When a 2 kg mass is attached to a vertical spring, the spring is stretched 10 cm such that the mass is 50 cm above the table.

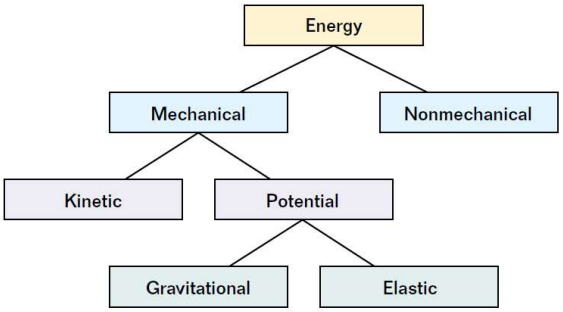
a) What is the gravitational potential energy associated with this mass relative to the table?

b) What is the spring's elastic potential energy if the spring constant is 400 N/m?

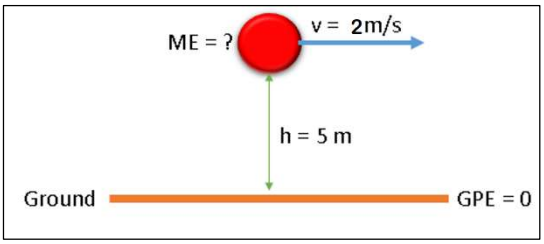
c) What is the total potential energy of this system?

Mechanical energy

The sum of kinetic energy and of all forms of potential energy is defined as mechanical energy: ME = KE + ΣPE



App14. A basketball of mass 0.625 kg is moving at 2 m/s at 5m above the ground. Find the basketball mechanical energy.



Conservation of mechanical energy

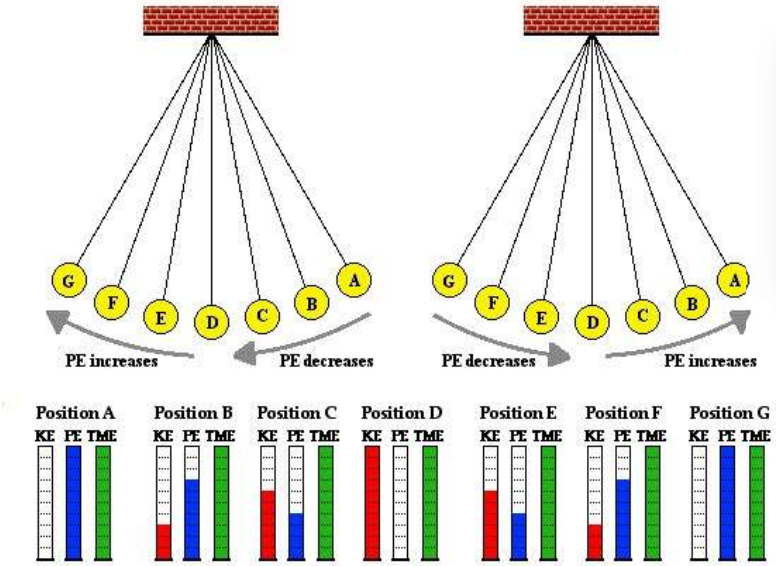
In absence of friction a system's mechanical energy will remain unchanged, that is potential can be converted into kinetic energy and vice versa, but their sum keeps same:

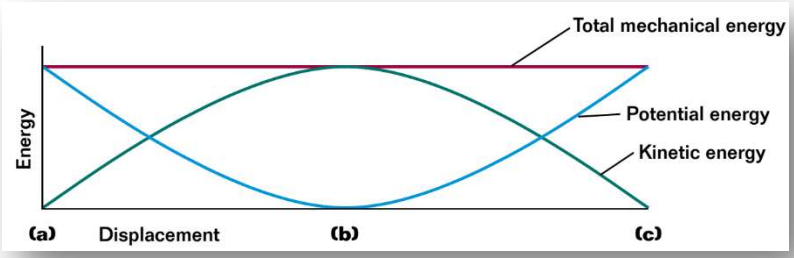
In the absence of friction: initial mechanical energy = final mechanical energy

$$ME_i = ME_f$$

$$KE_i + PE_i = KE_f + PE_f$$

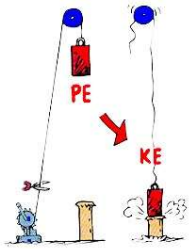
$$\frac{1}{2}mv_i^2 + mgh_i = \frac{1}{2}mv_f^2 + mgh_f$$



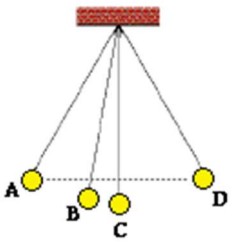


As the pendulum moves downward, its PE decreases while its KE increases. However, when it moves upward, its PE increases but its KE decreases. In either case, the sum of PE and KE (i.e., ME) remains constant (since friction is negligible).

App15. A 5kg mass hanging 10 m high from a rope (right figure). If the rope is cut, what speed does the mass hit the ground with?



App16. Find the missing energies for the pendulum in the next figure.



A: KE = 0 J
PE = 2.4 J

B: KE = 2.0 J
PE = _____ J

C: KE = _____ J
PE = 0 J

D: KE = _____ J
PE = _____ J

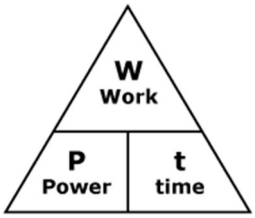
App17. A tennis ball thrown from ground level vertically upward at a speed of 10 m/s. Use the conservation of ME to find the maximum height it can reach.

App18. Starting from rest, a child zooms down a frictionless slide from an initial height of 3 m. What is her speed at the bottom of the slide? Assume the child has a mass of 25 kg.

App19. A pendulum bob is released from some initial height such that the speed of the bob at the bottom of the swing is 1.9 m/s. What is the initial height of the bob?

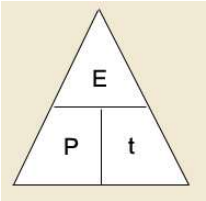
Power

- Notice that time wasn't yet mentioned in the previous calculations.
- The rate at which work is done or the rate at which energy is transferred is called power:



Average power = work ÷ time taken or $P_{av} = \frac{w}{\Delta t}$ $\quad or \quad P_{av} = \frac{En.transferred}{duration}$

- Since work is measured in joules (J) and time is seconds (s) then power is measured in (J/s),
  which is abbreviated watts (W).



- Note:

$$P_{av} = \frac{w}{\Delta t} = \frac{f.d}{\Delta t} = f.v_{av}$$

App20. An electric light bulb converts 100 J of electric energy into light and heat every 1min. Find its power.

App21. How long does it take a 19-kW steam engine to do 6.8 x 10^7 J of work?

App22. A 50 kg student climbs 5 m up a rope at a constant speed in 12.3 s.

a) How much work does the student do?

b) Find the student's power output.